

Electrically Charged Vortices in Non-Abelian Gauge Theories with Chern-Simons Term

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It is shown that a non-Abelian gauge theory with Higgs fields and the addition of a Chern-Simons term in 2+1 space-time dimensions exhibits finite-energy electrically charged vortex solutions. A novel feature of the vortices is that their electric charge Q is quantized in units of the fundamental charge e , $Q/e = n/2$ with n an integer, and their angular momentum is $J = Q/2e = n/4$.

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Gauge theories exhibit a rich spectrum of finite-energy (or finite-action) classical solutions. Vortices, monopoles, and instantons are the best known topological solutions in 2, 3, and 4 space dimensions, respectively.¹ Magnetic monopoles admit an electrically charged generalization in 3+1 space-time dimensions: dyons. Vortices, in both Abelian² and non-Abelian Higgs models,³⁻⁵ do not admit finite-energy charged generalizations in 2+1 dimensions.^{6,7}

The addition of a Chern-Simons (CS) term⁸⁻¹⁰ to the Abelian Higgs model (in 2+1 dimensions) changes this situation.¹¹ In fact, as a result of the presence of this term, vortices acquire electric charge keeping finite energy. An Abelian vortex with k units of magnetic flux has an electric charge $(2\pi\mu/e)k$, where μ , which has the dimension of a mass, is the coefficient of the CS term and e is the fundamental charge unit.

We present in this note finite-energy charged solutions for a non-Abelian Higgs model with a CS term in 2+1 dimensions. In this case, the CS term is topological in character and noninvariant under "large" gauge transformations, this last implying that μ must be quantized.¹⁰ As we shall see, this results in the quanti-

zation of the vortex electric charge Q which turns to be $Q = (e/2)n$ with n an integer; as a consequence, the associated angular momentum of the vortex is $J = n/4$.

Vortex solutions are associated with spontaneously broken gauge symmetries via Higgs fields. In order to have topologically stable vortices, the relevant homotopy group, $\pi_1(G/H)$, must be nontrivial. (G stands for the gauge group and H for the invariance group of the vacuum.) For $G = \text{SU}(N)$ and the Higgs fields in the adjoint representation it is convenient to have maximum symmetry breaking of G so that the vacuum is only invariant under the unit matrix in the adjoint representation. Then $H = Z_N$, $\pi_1(\text{SU}(N)/Z_N) = Z_N$, and we have $N-1$ topologically nontrivial homotopy classes besides the ordinary vacuum (trivial class).

Although the arguments that follow are general and apply to "realistic" theories such as the $\text{SU}(5)$ grand unified theory, it is useful to consider the simplest theory that admits non-Abelian vortices.³⁻⁵ This is the theory of a gauge field A_μ taking values in the Lie algebra of $\text{SU}(2)$ (with generators t^a) coupled to two Higgs fields (in order to achieve maximum symmetry breaking) in the adjoint representation. The Lagrangian density (in 2+1 space-time dimensions) reads

$$\mathcal{L} = -\frac{1}{4}\mathbf{F}_{\mu\nu} \cdot \mathbf{F}^{\mu\nu} + \frac{1}{2}D_\mu\phi \cdot D^\mu\phi + \frac{1}{2}D_\mu\psi \cdot D^\mu\psi + \frac{1}{4}\mu\epsilon_{\mu\nu\alpha}[\mathbf{F}^{\alpha\mu} \cdot \mathbf{A}^\nu - \frac{2}{3}e\mathbf{A}^\alpha \cdot (\mathbf{A}^\mu \times \mathbf{A}^\nu)] - V(\phi, \psi) \quad (1)$$

with $A_\mu = A_\mu^a t^a$ and

$$D_\mu\phi = \partial_\mu\phi + e\mathbf{A}_\mu \times \phi, \quad \mathbf{F}_{\mu\nu} = \partial_\mu\mathbf{A}_\nu - \partial_\nu\mathbf{A}_\mu + e\mathbf{A}_\mu \times \mathbf{A}_\nu.$$

The fourth term on the right-hand side of Eq. (1) is the Chern-Simons term. It violates both P and T invariance but not C invariance. Although it leads to gauge-covariant equations of motion, it is not itself gauge invariant; rather, it changes by a total derivative. It then follows that the response of the action to a gauge transformation u is

$$S = \int d^3x \mathcal{L} \rightarrow \int d^3x \mathcal{L} + \mu(8\pi^2/e^2)\omega(u),$$

where $\omega(u)$ is the winding number of the gauge transformation. In order to make $\exp(iS)$ gauge invariant, one then has to impose

$$\mu = -e^2 n/4\pi, \quad n \in \mathbb{Z}. \quad (2)$$

For $G = \text{SU}(2)$ there is only one topologically stable vortex with ϕ and ψ not parallel in isospace at the minimum of the potential $V(\phi, \psi)$ which can be taken, for example, as

$$V(\phi, \psi) = \frac{1}{8}g^2(\phi^2 - \eta^2)^2 + \frac{1}{8}g'^2(\psi^2 - \eta'^2)^2 + \frac{1}{2}g''^2(\psi \cdot \phi)^2. \quad (3)$$

We make the following *Ansatz* for the charged vortex solution:

$$\begin{aligned}\phi &= j(\rho)(\cos\varphi, \sin\varphi, 0), \quad \psi = \hat{e}_3\eta', \\ \mathbf{A}_\varphi &= \hat{e}_3 A(\rho), \quad \mathbf{A}_0 = \hat{e}_3 A_0(\rho)\end{aligned}\quad (4)$$

with $\hat{e}_3 = (0, 0, 1)$ and (ρ, φ) polar coordinates. One can also make another vortex *Ansatz* exchanging the coordinate dependence of ϕ and ψ . It can be seen that Eq. (4) leads to the lowest vortex solution provided $\eta < \eta'$ and $g < g'$ or $m < m'$ and $g > g'$ (where m is the Higgs mass, $m = \eta g$).

Finite action requires the following boundary conditions:

$$A(\infty) = -1/e, \quad A_0(\infty) = 0, \quad f(\infty) = \eta,$$

and

$$A(0) = A_0(0) = f(0) = 0.$$

In order to define an electromagnetic field $\mathcal{F}_{\mu\nu}$ we have at hand two Higgs fields. However, our *Ansatz* corresponds to $\mathbf{F}_{\mu\nu}$ and ϕ mutually orthogonal. We

hence propose

$$\mathcal{F}_{\mu\nu} = (\psi/|\psi|) \cdot \mathbf{F}_{\mu\nu} = F_{\mu\nu}^3. \quad (5)$$

The electric and magnetic fields take then the form

$$E_i = \mathcal{F}_{0i}, \quad B = \frac{1}{2}\epsilon_{ij}\mathcal{F}_{ij}. \quad (6)$$

The topological charge k of the vortex configuration is related to the magnetic flux: $k = e\Phi/2\pi$ with $\Phi = \int d^2x B$. From the Stokes theorem and the boundary conditions one gets

$$\Phi = -2\pi/e \quad (7)$$

and hence $k = -1$ for our solution. With *Ansatz* (4) the equations of motion

$$\begin{aligned}D_\mu D^\mu \phi &= -\frac{\delta V}{\delta \phi}, \quad D_\mu D^\mu \psi = -\frac{\delta V}{\delta \psi}, \\ D_\mu \mathbf{F}^{\mu\nu} &= \mathbf{j}^\nu + \frac{1}{2}\mu\epsilon^{\nu\alpha\beta}\mathbf{F}_{\alpha\beta},\end{aligned}\quad (8)$$

$$\mathbf{j}^\nu \equiv e(D^\nu \phi \times \phi + D^\nu \psi \times \psi)$$

become

$$\begin{aligned}A_0''(\rho) + \rho^{-1}A_0'(\rho) - e^2 f^2 A_0(\rho) &= -(\mu/\rho)A'(\rho), \quad A''(\rho) - \rho^{-1}A'(\rho) - ef^2(1 + eA) = -\mu\rho A_0'(\rho), \\ f''(\rho) + \rho^{-1}f'(\rho) - \rho^{-2}(1 + eA)^2/f(\rho) + e^2 A_0^2 f &= \frac{1}{2}g^2(f^2 - \eta^2)f.\end{aligned}\quad (9)$$

As a result of the presence of the CS term only charged solutions are possible since Eq. (9) leads to a trivial $\mathbf{A}$ for $A_0 = 0$. One can easily analyze the asymptotic behavior of the charged solution from Eqs. (9). One finds two possible solutions for large ρ :

$$\begin{aligned}A(\rho) &= -1/e + Z_\pm(m\rho/e)K_1(m_\pm\rho)[1 + O(e^{-m_\pm\rho})], \quad A_0(\rho) = \pm Z_\pm(m/e)K_0(m_\pm\rho)[1 + O(e^{-m_\pm\rho})], \\ f(\rho) &= \eta[1 - Y_\pm K_0(m\rho) + O(e^{-2m\rho})],\end{aligned}\quad (10)$$

where $Z_\pm$, $Y_\pm$ are dimensionless constants and $m_\pm$ the two distinct vector meson masses,

$$m_\pm = (\frac{1}{4}\mu^2 + e^2\eta^2)^{1/2} \pm \frac{1}{2}\mu. \quad (11)$$

Concerning the electric charge of the vortex configuration, note that the first of Eqs. (9) can be written in the form

$$\partial^i E_i + \mu B = \sigma, \quad (12)$$

where the charge density σ reads

$$\sigma \equiv \hat{e}_3 \cdot \mathbf{J}_0 = e^2 f^2 A_0.$$

Since $\lim_{\rho \rightarrow \infty} E_i = 0$ one gets from Eq. (12) a relation between the electric charge Q ,

$$Q = \int d^2x \sigma, \quad (13)$$

and the magnetic flux:

$$Q = \mu\Phi. \quad (14)$$

This important relation, already recognized in Ref. 10, is also valid in the Abelian model. In the present non-Abelian case, it implies a quantization condition

for Q . Indeed, from Eqs. (2), (7), and (14) we get

$$Q = (e/2)n, \quad n \in \mathbb{Z}. \quad (15)$$

The charge quantization can be connected with the angular momentum of the vortex. In two space dimensions there is only one generator of angular momentum:

$$J = \int d^2x \epsilon^{ij} x_i T_{0j},$$

where $T_{\mu\nu}$ is the energy-momentum tensor. For the vortex solution, the only nontrivial momentum density is $T_{0\varphi}$. Using its explicit form one gets

$$\begin{aligned}J &= 2\pi \int_0^\infty \rho d\rho e A_0 f^2 + \frac{1}{2}\mu\pi A(\infty)^2 \\ &= Q/e + \pi\mu/e^2\end{aligned}$$

or

$$J = Q/2e.$$

Hence, the vortex carries a quantized angular momentum, $J = n/4$ (in contrast with the Abelian case where, μ not being quantized, it can take any continuum

value).

As is well known, vortices exist provided a Ginzburg-Landau-type parameter λ , which in the Higgs model is related to the scalar and vector meson masses,

$$\lambda = m/m_{\text{vector}}$$

satisfies the condition $\lambda > 1$ corresponding to type-II superconductivity. This leads to

$$\mu = ne^2/4\pi > g\eta(e^2/g^2 - 1)$$

for $m_{\text{vector}} = m_+$ and

$$0 < -ne^2/4\pi < g\eta(e^2/g^2 - 1)$$

for the m_- solution (this is also valid in the Abelian model¹¹). Concerning the energy ϵ of the vortex solution, from the asymptotic fields, Eq. (10), it follows that $\epsilon \sim m_{\pm}^2 \ln(e^2/m_{\pm}^2)$ and hence the m_- solution has lower energy.

Qualitatively, the field behavior is the following: The magnetic field decreases monotonically from its value at the origin to zero at infinity with characteristic length $1/m_-$; the scalar field ϕ increases with characteristic length $1/m$ from zero at the origin to its vacuum value at infinity. Finally, the electric field vanishes both at the origin and at infinity, reaching its maximum at some finite ρ .

As we stated above, an $SU(N)$ theory admits vortices with topological charge $k = 1, 2, \dots, N-1$. In that case, the charge $Q = (en/2\sqrt{N})a$ with $a = 1, 2, \dots, N-1$.

We conclude by noting that (Euclidean) three-dimensional field theories can be used to study four-dimensional physics at high temperatures. In this context, charged vortex lines can find important applications in the early universe.

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